

BANK ATM MANAGER



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Roll Number:

BSCS25109

Degree:

BS Computer Science (Semester I)

Section:

A

Subject:

Programming Fundamental

PROJECT:

01

```

/*
account.txt sample data:
accountNumber  pin  balance  status  failedAttempts  locked  needsApproval
10001          1122  5000    A       0              0       0
10002          3344  2000    A       0              0       0
10003          5566  0       A       0              0       1
10004          7788  15000   D       0              0       0
...

transactions.txt sample data:
TransactionID  Type  SourceAccount  DestinationAccount  Amount
1              1     10001          -1                  2000
2              4     10001          -1                  1000
3              2     10001          10002              500
4              3     10002          10001              500
...

admin.txt sample data:
GXHX7:65>      ->BSCS25109
*/

```

```

#include <iostream>
#include <iomanip>
#include <fstream>
#include <cstdlib>
#include <conio.h>
#include <windows.h>
using namespace std;

const int maxAccounts = 2000;
const int loginAttempts = 3;

long long accountNumbers[maxAccounts];
char pins[maxAccounts][6];
long long balances[maxAccounts];
char status[maxAccounts];
int failedAttempts[maxAccounts];
int locked[maxAccounts];
int totalAccounts = 0;
int nextTransactionID = 1;
int needsApproval[maxAccounts];

void saveAccounts()
{
    ofstream fout("accounts.txt");
    if(!fout)
    {
        cout << "No accounts file found"<<endl;
    }
}

```

```

46 {
47     ofstream fout("accounts.txt");
48     if(!fout)
49     {
50         cout << "No accounts file found"<<endl;
51         return;
52     }
53     for (int i = 0; i < totalAccounts; i++)
54     {
55         fout << accountNumbers[i] << " " << pins[i] << " " << balances[i] << " " << status[i] << " " << failedAttempts[i] << " " << locked[i] << " " << needsApproval[i] << endl;
56     }
57     fout.close();
58 }
59
60
61 void loadAccounts()
62 {
63     ifstream fin("accounts.txt");
64     if (!fin)
65     {
66         cout << "No accounts file found"<<endl;
67         return;
68     }
69     totalAccounts = 0;
70     while (fin >> accountNumbers[totalAccounts] >> pins[totalAccounts] >> balances[totalAccounts] >> status[totalAccounts] >> failedAttempts[totalAccounts] >> locked[totalAccounts] >> needsApproval[totalAccounts])
71     {
72         totalAccounts++;
73     }
74     fin.close();
75 }
76
77
78 void recordTransactions(int type, long long source, long long destination, long long amount)
79 {
80     ifstream fin("transactions.txt");
81     int lastID = 0;
82     int tranID;
83     int t;
84     long long s;
85     int d;
86     int a;
87     while (fin >> tranID >> t >> s >> d >> a)
88     {
89         lastID = tranID;
90     }
91     fin.close();
92     int nextID = lastID + 1;
93     ofstream fout("transactions.txt", ios::app);
94     fout << nextID << " " << type << " " << source << " " << destination << " " << amount << endl;
95     fout.close();
96 }

```

```

104
105     void deposit(int index)
106     {
107         long long amount;
108         cout << "Enter amount to deposit: ";
109         cin >> amount;
110
111         if (amount <= 0)
112         {
113             cout << "Invalid amount"<<endl;
114             return;
115         }
116
117         balances[index] = balances[index]+ amount;
118
119         recordTransactions(1, accountNumbers[index], -1, amount);
120         saveAccounts();
121
122         cout << "Deposit successful"<<endl<<endl;
123         cout << "New balance : " << balances[index] << endl;
124     }
125
126
127     void withdraw(int index)
128     {
129         long long amount;
130         cout << "Enter amount to withdraw: ";
131         cin >> amount;
132         if (amount <= 0 || amount > balances[index])
133         {
134             cout << "Invalid or insufficient balance"<<endl;
135             return;
136         }
137
138         balances[index] = balances[index] - amount;
139
140         recordTransactions( 4, accountNumbers[index], -1, amount);
141         saveAccounts();
142
143         cout << "Withdraw successful" << endl<<endl;
144         cout << "New balance : " << balances[index] << endl;
145     }
146
147     void transfer(int index)
148     {
149         long long destination;
150         long long amount;
151         int destIdx = -1;
152         cout << "Enter destination account: ";
153         cin >> destination;
154         cout << "Enter amount to transfer: ";
155         cin >> amount;
156

```

```

157
158     for (int i = 0; i < totalAccounts; i++)
159     {
160         if (accountNumbers[i] == destination)
161         {
162             destIdx = i;
163             break;
164         }
165     }
166
167     if (destIdx == -1)
168     {
169         cout << "Destination account does not exist. Check the number and try again." << endl;
170         return;
171     }
172     if (amount <= 0)
173     {
174         cout << "Invalid transfer amount. Must be greater than zero." << endl;
175         return;
176     }
177     if (amount > balances[index])
178     {
179         cout << "Insufficient balance. Your current balance is " << balances[index] << endl;
180         return;
181     }
182
183     balances[index] = balances[index] - amount;
184     balances[destIdx] = balances[destIdx] + amount;
185
186     recordTransactions(2, accountNumbers[index], destination, amount);
187     recordTransactions(3, destination, accountNumbers[index], amount);
188
189     saveAccounts();
190
191     cout << "Transfer successful" << endl;
192 }
193
194
195
196
197 void checkBalance(int index)
198 {
199     cout << "Current Balance: " << balances[index] << endl;
200 }
201

```

```

201
202 void miniStatement(long long accountNo)
203 {
204     ifstream fin("transactions.txt");
205     if (!fin)
206     {
207         cout << "No transactions found" << endl;
208         return;
209     }
210
211     int ids[1000];
212     int types[1000];
213     long long sources[1000];
214     long long destinations[1000];
215     long long amounts[1000];
216     int count = 0;
217
218
219     while (fin >> ids[count] >> types[count] >> sources[count] >> destinations[count] >> amounts[count])
220     {
221         count++;
222         if (count >= 1000)
223         {
224             break;
225         }
226     }
227     fin.close();
228
229
230     int k;
231     cout << "Enter number of last transactions to view: ";
232     cin >> k;
233     if (k > count)
234     {
235         k = count;
236     }
237     cout << "<=====><0.0><=====>" << endl;
238     cout << "ID      Type      Amount      Counterparty" << endl;
239     cout << "<=====><0.0><=====>" << endl;
240
241     int shown = 0;
242     for (int i = count - 1; i >= 0 && shown < k; i--)
243     {
244         if (sources[i] == accountNo || destinations[i] == accountNo)
245         {
246             cout << setw(3) << ids[i] << " ";
247
248             if (types[i] == 1)
249             {
250                 cout << "Deposit      ";
251             }
252             else if (types[i] == 2)
253             {
254                 cout << "TransferOut ";

```

```

249         {
250             cout << "Deposit      ";
251         }
252     else if (types[i] == 2)
253     {
254         cout << "TransferOut  ";
255     }
256     else if (types[i] == 3)
257     {
258         cout << "TransferIn   ";
259     }
260     else if (types[i] == 4)
261     {
262         cout << "Withdraw     ";
263     }
264     else
265     {
266         cout << "Unknown      ";
267     }
268
269     cout << setw(8) << amounts[i] << " ";
270
271     if (types[i] == 2)
272     {
273         cout << " " << amounts[i];
274     }
275     else if (types[i] == 3)
276     {
277         cout << " " << sources[i];
278     }
279
280     cout << endl;
281     shown++;
282 }
283
284
285 if (shown == 0)
286 {
287     cout << "No transactions found for this account"<<endl;
288 }
289
290 void changePIN(int index)
291 {
292     cout << "Enter current PIN: ";
293     char oldPIN[6];
294     char ch;
295     int i = 0;
296     while ((ch = _getch()) != 13 && i < 5)
297     {
298         oldPIN[i] = ch;
299         cout << '*';
300         i++;
301     }
302     oldPIN[i] = '\0';

```

```

301     }
302     oldPIN[i] = '\0';
303     cout << endl;
304
305     int match = 1;
306     for (int j = 0; j < 6; j++)
307     {
308         if (pins[index][j] != oldPIN[j])
309         {
310             match = 0;
311             break;
312         }
313         if (pins[index][j] == '\0' && oldPIN[j] == '\0')
314         {
315             break;
316         }
317     }
318     if (!match)
319     {
320         cout << "Incorrect PIN"<<endl;
321         {
322             return;
323         }
324     }
325
326     cout << "Enter new PIN: ";
327     char newpin1[6];
328     i = 0;
329     while ((ch = _getch()) != 13 && i < 4)
330     {
331         newpin1[i++] = ch;
332         cout << '*';
333     }
334     newpin1[i] = '\0';
335     cout << endl;
336
337     cout << "Please Re enter new PIN again to confirm : ";
338     char newpin2[6];
339     i = 0;
340     while ((ch = _getch()) != 13 && i < 4)
341     {
342         newpin2[i] = ch;
343         cout << '*';
344         i++;
345     }
346     newpin2[i] = '\0';
347     cout << endl;
348
349
350
351     int len = 0;
352     while (newpin1[len] != '\0')
353     {
354         len++;
355     }

```



```

352     while (newpin1[len] != '\0')
353     {
354         len++;
355     }
356     if (len != 4)
357     {
358         cout << "PIN must be exactly four digits"<<endl;
359         return;
360     }
361
362     for (int j = 0; j < 4; j++)
363     {
364         if (newpin1[j] < '0' || newpin1[j] > '9')
365         {
366             cout << "PIN must contain only digits"<<endl;
367             return;
368         }
369     }
370     int same = 1;
371     for (int j = 0; j < 6; j++)
372     {
373         if (newpin1[j] != newpin2[j])
374         {
375             same = 0;
376             break;
377         }
378         if (newpin1[j] == '\0' && newpin2[j] == '\0')
379         {
380             break;
381         }
382     }
383     if (!same)
384     {
385         cout << "PIN mismatch please try again after some time"<<endl;
386         {
387             return;
388         }
389     }
390     for (int j = 0; j < 6; j++)
391     {
392         pins[index][j] = newpin1[j];
393     }
394     saveAccounts();
395
396     cout << "PIN changed successfully"<<endl;
397 }
398
399
400 void customerMenu(int index)
401 {
402     int c;
403     do {
404         system("cls");
405         cout << "<=====>" << endl;
406         cout << "          CUSTOMER MENU          " << endl;
407         cout << "<=====>" << endl;

```

```

409         cout << "2. Withdraw" << endl;
410         cout << "3. Check Balance" << endl;
411         cout << "4. Transfer" << endl;
412         cout << "5. Change PIN" << endl;
413         cout << "6. Mini Statement" << endl;
414         cout << "7. Logout" << endl<<endl;
415         cout << "<----->" << endl;
416         cout << "Enter choice: ";
417         cin >> c;
418
419         if (c == 1)
420         {
421             deposit(index);
422         }
423         else if (c == 2)
424         {
425             withdraw(index);
426         }
427         else if (c == 3)
428         {
429             checkBalance(index);
430         }
431         else if (c == 4)
432         {
433             transfer(index);
434         }
435         else if (c == 5)
436         {
437             changePIN(index);
438         }
439         else if (c == 6)
440         {
441             miniStatement(accountNumbers[index]);
442         }
443
444         if (c != 7)
445         {
446             cout <<endl<< "Press any key to continue";
447             _getch();
448         }
449     } while (c != 7);
450 }
451
452
453 void customerLogin()
454 {
455     long long account;
456     int index = 0;
457     cout << "Enter account number: ";
458     cin >> account;
459
460     for (int i = 0; i < totalAccounts; i++)
461     {
462         if (accountNumbers[i] == account)
463         {
464             index = i;

```

```

466     }
467
468     if (index == -1)
469     {
470         cout << "Account not found"<<endl;
471         return;
472     }
473     if (status[index] != 'A')
474     {
475         cout << "Account not active"<<endl;
476         return;
477     }
478     if (locked[index])
479     {
480         cout << "Account locked"<<endl;
481         return;
482     }
483
484     int attempts = 0;
485     while (attempts < loginAttempts)
486     {
487         cout << "Enter PIN: ";
488
489         char pin[6];
490         char ch;
491         int i = 0;
492         while ((ch = _getch()) != 13 && i < 5)
493         {
494             pin[i] = ch;
495             cout << '*';
496             i++;
497         }
498         pin[i] = '\0';
499         cout << endl;
500
501         int found = 1;
502         for (int j = 0; j < 6; j++)
503         {
504             if (pins[index][j] != pin[j])
505             {
506                 found = 0;
507                 break;
508             }
509             if (pins[index][j] == '\0' && pin[j] == '\0')
510             {
511                 break;
512             }
513         }
514
515         if (found)
516         {
517             failedAttempts[index] = 0;
518             saveAccounts();
519
520             if (needsApproval[index])
521             {

```

```

529         customerMenu(index);
530         return;
531     }
532     else
533     {
534         attempts++;
535         failedAttempts[index]++;
536         cout << "Incorrect PIN. Attempts left: " << (3 - attempts) << endl;
537     }
538 }
539 locked[index] = 1;
540 saveAccounts();
541 cout << "Account locked due to 3 failed attempts."<<endl;
542 }
543
544
545
546 void openAccount()
547 {
548     long long account;
549
550     if (totalAccounts == 0)
551     {
552         account = 10001;
553     }
554     else
555     {
556         account = accountNumbers[totalAccounts - 1] + 1;
557     }
558
559     cout << "New Account Number assigned: " << account << endl;
560
561
562     char pin[6];
563     int pin2[6];
564     char ch;
565     int i;
566     do
567     {
568         cout << "Enter 4-digit PIN: ";
569         i = 0;
570         while ((ch = _getch()) != 13 && i < 4)
571         {
572             if (ch >= '0' && ch <= '9')
573             {
574                 pin[i] = ch;
575                 cout << '*';
576                 i++;
577             }
578         }
579         pin[i] = '\0';
580         cout << endl;
581
582         if (i != 4)
583         {
584             cout << "PIN must be exactly 4 digits, Try again"<<endl;

```

```

580     cout << endl;
581
582     if (i != 4)
583     {
584         cout << "PIN must be exactly 4 digits, Try again"<<endl;
585     }
586     while (i != 4);
587
588     i = 0;
589     cout << "Re-enter PIN: ";
590     while ((ch = _getch()) != 13 && i < 4)
591     {
592         if (ch >= '0' && ch <= '9')
593         {
594             pin2[i] = ch;
595             cout << '*';
596             i++;
597         }
598     }
599     pin2[i] = '\0';
600     cout << endl;
601
602
603     int same = 1;
604     for (int j = 0; j < 5; j++)
605     {
606         if (pin[j] != pin2[j])
607         {
608             same = 0;
609             break;
610         }
611         if (pin[j] == '\0' && pin2[j] == '\0')
612             break;
613     }
614
615     if (!same)
616     {
617         cout << "PIN mismatch. Account creation cancel"<<endl;
618         return;
619     }
620
621     for (int j = 0; j < 6; j++)
622     {
623         pins[totalAccounts][j] = pin[j];
624     }
625
626
627     long long deposit;
628     do
629     {
630         cout << "Enter initial deposit : ";
631         cin >> deposit;
632         if (deposit < 0)
633         {
634             cout << "Deposit cannot be negative"<<endl;
635         }

```

```

634         cout << "Deposit cannot be negative"<<endl;
635     }
636 } while (deposit < 0);
637
638 balances[totalAccounts] = deposit;
639
640 accountNumbers[totalAccounts] = account;
641 status[totalAccounts] = 'A';
642 failedAttempts[totalAccounts] = 0;
643 locked[totalAccounts] = 0;
644 needsApproval[totalAccounts] = 1;
645
646 totalAccounts++;
647 saveAccounts();
648
649 cout << "Account created successfully" << endl << endl;
650 cout << "New Account Number is: " << account << endl;
651 }
652
653 void closeAccount()
654 {
655     long long account;
656     cout << "Enter Account Number to close: ";
657
658     cin >> account;
659     for (int i = 0; i < totalAccounts; i++)
660     {
661         if (accountNumbers[i] == account)
662         {
663             status[i] = 'C';
664             saveAccounts();
665             cout << "Account closed" << endl;
666             return;
667         }
668     }
669     cout << "Account not found" << endl;
670 }
671
672 void changeAccountStatus()
673 {
674     long long account;
675     cout << "Enter Account Number: ";
676     cin >> account;
677     for (int i = 0; i < totalAccounts; i++)
678     {
679         if (accountNumbers[i] == account)
680         {
681             if (status[i] == 'A')
682             {
683                 status[i] = 'D';
684             }
685             else if (status[i] == 'D')
686             {
687                 status[i] = 'A';
688             }
689         }

```

```

689         saveAccounts();
690         cout << "Account status changed" << endl;
691         return;
692     }
693 }
694 }
695 cout << "Account not found" << endl;
696 }
697
698 void unlockAccount()
699 {
700     long long account;
701     cout << "Enter Account Number to unlock: ";
702     cin >> account;
703     for (int i = 0; i < totalAccounts; i++)
704     {
705         if (accountNumbers[i] == account)
706         {
707             locked[i] = 0;
708             failedAttempts[i] = 0;
709             saveAccounts();
710             cout << "Account unlocked" << endl;
711             return;
712         }
713     }
714     cout << "Account not found" << endl;
715 }
716
717 void searchAccount()
718 {
719     long long account;
720     cout << "Enter Account Number: ";
721     cin >> account;
722     for (int i = 0; i < totalAccounts; i++)
723     {
724         if (accountNumbers[i] == account)
725         {
726             cout << "Account: " << accountNumbers[i] << " | Bal: " << balances[i] << " | Status: " << status[i] << endl;
727             return;
728         }
729     }
730     cout << "Not found" << endl;
731 }
732 void swapAccounts(int i, int j)
733 {
734     long long tempAcc;
735     long long tempBal;
736     char tempStatus;
737     int tempLocked;
738
739     tempAcc = accountNumbers[i];
740     accountNumbers[i] = accountNumbers[j];
741     accountNumbers[j] = tempAcc;
742
743     tempBal = balances[i];
744     balances[i] = balances[j];
745     balances[j] = tempBal;
746
747     tempStatus = status[i];
748     status[i] = status[j];
749     status[j] = tempStatus;
750
751     tempLocked = locked[i];
752     locked[i] = locked[j];
753     locked[j] = tempLocked;
754
755     saveAccounts();
756 }

```

```

736     char tempStatus;
737     int tempLocked;
738
739
740     tempAcc = accountNumbers[i];
741     accountNumbers[i] = accountNumbers[j];
742     accountNumbers[j] = tempAcc;
743
744
745     tempBal = balances[i];
746     balances[i] = balances[j];
747     balances[j] = tempBal;
748
749
750     tempStatus = status[i];
751     status[i] = status[j];
752     status[j] = tempStatus;
753
754
755     tempLocked = locked[i];
756     locked[i] = locked[j];
757     locked[j] = tempLocked;
758 }
759
760 void displayAllAccounts()
761 {
762     if (totalAccounts == 0)
763     {
764         cout << "No accounts to display" << endl;
765         return;
766     }
767
768     int order;
769     cout << "=====>" << endl;
770     cout << "    Display All Accounts    " << endl;
771     cout << "=====>" << endl;
772     cout << " 1. Ascending " << endl;
773     cout << " 2. Descending "<<endl;
774     cout << "Enter choice: ";
775     cin >> order;
776
777     if (totalAccounts <= 1)
778     {
779         cout << "Nothing to sort. Only " << totalAccounts << " account available." << endl;
780         return;
781     }
782     for (int i = 0; i < totalAccounts - 1; i++)
783     {
784         for (int j = 0; j < totalAccounts - i - 1; j++)
785         {
786             bool swapFlag = false;
787
788             if ((order == 1 && balances[j] > balances[j + 1]) || (order == 2 && balances[j] < balances[j + 1]))
789             {
790                 swapFlag = true;

```



```

778     {
779         cout << "Nothing to sort. Only " << totalAccounts << " account available." << endl;
780         return;
781     }
782     for (int i = 0; i < totalAccounts - 1; i++)
783     {
784         for (int j = 0; j < totalAccounts - i - 1; j++)
785         {
786             bool swapFlag = false;
787
788             if ((order == 1 && balances[j] > balances[j + 1]) || (order == 2 && balances[j] < balances[j + 1]))
789             {
790                 swapFlag = true;
791             }
792
793             if (swapFlag)
794             {
795                 swapAccounts(j, j + 1);
796             }
797         }
798     }
799
800     cout << "AccNo\tBalance\tStatus\tLocked" << endl;
801     for (int i = 0; i < totalAccounts; i++)
802     {
803         cout << accountNumbers[i] << "\t" << balances[i] << "\t" << status[i] << "\t" << locked[i] << endl;
804     }
805 }
806
807
808
809 void summaryReport()
810 {
811     int active = 0;
812     int closed = 0;
813     int deactive = 0;
814     int lockcount = 0;
815     long long totalBalance = 0;
816     for (int i = 0; i < totalAccounts; i++)
817     {
818         if (status[i] == 'A')
819         {
820             active++;
821         }
822         if (status[i] == 'C')
823         {
824             closed++;
825         }
826         if (status[i] == 'D')
827         {
828             deactive++;
829         }
830         if (locked[i] == 1)
831         {
832             lockcount++;
833         }
834     }

```

```

833     }
834     totalBalance += balances[i];
835 }
836
837 cout << "<=====>" << endl;
838 cout << "          Summary Report          " << endl;
839 cout << "<=====>" << endl;
840 cout << "1. Active: " << active << endl;
841 cout << "2. Closed: " << closed << endl;
842 cout << "3. Deactivated: " << deactivate << endl;
843 cout << "4 .Locked: " << lockcount << endl;
844 cout << "5. Total Balance: " << totalBalance << endl;
845 }
846
847 void approveAccounts()
848 {
849     cout << "<=====>" << endl;
850     cout << "          Accounts Pending Approval          " << endl;
851     cout << "<=====>" << endl;
852     bool found = false;
853     for (int i = 0; i < totalAccounts; i++)
854     {
855         if (needsApproval[i])
856         {
857             cout << "Account: " << accountNumbers[i] << endl;
858             cout << "Balance: " << balances[i] << endl;
859             found = true;
860         }
861     }
862     if (!found)
863     {
864         cout << "No accounts pending approval" << endl;
865         return;
866     }
867     long long account;
868     cout << "Enter account number to approve : ";
869
870     cin >> account;
871
872
873     for (int i = 0; i < totalAccounts; i++)
874     {
875         if (accountNumbers[i] == account)
876         {
877             needsApproval[i] = 0;
878             saveAccounts();
879             cout << "Account " << account << " approved successfully" << endl;
880             return;
881         }
882     }
883
884     cout << "Account not found" << endl;;
885 }
886

```

```

887
888
889 void managerMenu()
890 {
891     int ch;
892     do
893     {
894         system("cls");
895         cout << "<=====>" << endl;
896         cout << "          MANAGER MENU          " << endl;
897         cout << "<=====>" << endl;
898         cout << "1. Open New Account" << endl;
899         cout << "2. Close Account" << endl;
900         cout << "3. Deactivate/Activate" << endl;
901         cout << "4. Unlock Account" << endl;
902         cout << "5. Search Account" << endl;
903         cout << "6. Display All" << endl;
904         cout << "7. Summary Report" << endl;
905         cout << "8. Approve Accounts" << endl;
906         cout << "9. Back" << endl<<endl;
907         cout << "<=====>" << endl<<endl;
908         cout << "Enter choice: ";
909         cin >> ch;
910
911         if (ch == 1)
912         {
913             openAccount();
914         }
915         else if (ch == 2)
916         {
917             closeAccount();
918         }
919         else if (ch == 3)
920         {
921             changeAccountStatus();
922         }
923         else if (ch == 4)
924         {
925             unlockAccount();
926         }
927         else if (ch == 5)
928         {
929             searchAccount();
930         }
931         else if (ch == 6)
932         {
933             displayAllAccounts();
934         }
935         else if (ch == 7)
936         {
937             summaryReport();
938         }
939         else if (ch == 8)
940         {
941             approveAccounts();

```

```

942     }
943
944     if (ch != 9)
945     {
946         cout << endl << "Press any key to continue";
947         _getch();
948     }
949
950     } while (ch != 9);
951
952     }
953
954     int managerLogin()
955     {
956         ifstream fin("admin.txt");
957
958         if (!fin)
959         {
960             cout << "File not found" << endl;
961             return 0;
962         }
963         char encryptedPass[40];
964         char realPass[40];
965         char inputPass[40];
966         fin >> encryptedPass;
967         fin.close();
968
969         int i = 0;
970         while (encryptedPass[i] != '\0')
971         {
972             realPass[i] = encryptedPass[i] - 5;
973             i++;
974         }
975         realPass[i] = '\0';
976
977         cout << "Enter Manager Password: ";
978         char ch;
979         int j = 0;
980         while ((ch = _getch()) != 13 && j < 39)
981         {
982             inputPass[j] = ch;
983             cout << '*';
984             j++;
985         }
986         inputPass[j] = '\0';
987         cout << endl;
988
989         int match = 1;
990         for (int j = 0; realPass[j] != '\0' || inputPass[j] != '\0'; j++)
991         {
992             if (realPass[j] != inputPass[j])
993             {
994                 match = 0;

```

```

996     }
997 }
998
999 if (match)
1000 {
1001     cout << "Manager login successful" << endl;
1002     return 1;
1003 }
1004 else
1005 {
1006     cout << "Invalid manager password please try again " << endl;
1007     return 0;
1008 }
1009 }
1010
1011
1012 int main()
1013 {
1014     loadAccounts();
1015
1016     int ch;
1017     do
1018     {
1019         Sleep(1500);
1020         system("cls");
1021         cout << "<=====>" << endl;
1022         cout << "      BANK ATM SYSTEM      " << endl;
1023         cout << "<=====>" << endl;
1024         cout << "1. Customer Login" << endl;
1025         cout << "2. Manager Login" << endl;
1026         cout << "3. Exit" << endl << endl;
1027         cout << "<=====>" << endl << endl;
1028         cout << "Enter choice: ";
1029         cin >> ch;
1030
1031         if (ch == 1)
1032         {
1033             customerLogin();
1034         }
1035         else if (ch == 2)
1036         {
1037             if (managerLogin() == 1)
1038             {
1039                 managerMenu();
1040             }
1041         }
1042     } while (ch != 3);
1043
1044
1045

```

```

1045
1046     system("cls");
1047     cout << "<=====>" << endl;
1048     cout << "      Thank you for using BANK ATM SYSTEM      " << endl;
1049     cout << "          Your money is safe                      " << endl;
1050     cout << "                  /\_\_/\_\                      " << endl;
1051     cout << "                  ( o.o )                        " << endl;
1052     cout << "                  > ^ <                          " << endl;
1053     cout << "          Have a secure and pleasant day          " << endl;
1054     cout << "<=====>" << endl;
1055
1056     return 0;
1057 }
1058
1059

```

```
1      1 1 10001 -1 123
2      2 4 10001 -1 1232
3      3 2 10001 10003 100
4      4 3 10003 10001 100
5      |
```

```
accounts.txt  admin.txt  transactions.txt  bscs25109 PRO 01.cpp
1             GXHX7:65>
```

```
accounts.txt  admin.txt  transactions.txt  bscs25109 PRO 01.cpp
1             10001 1234 3791 A 0 0 0
2             10002 3344 2000 A 0 0 0
3             10003 5566 100 A 0 0 0
4             10004 7788 15000 A 0 0 0
5             10005 1111 1000 A 0 0 0
6             |
```

```
D:\All things\Abu Bakar\Bank
<=====>
      BANK ATM SYSTEM
<=====>
1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: 
```



D:\All things\Abu Bakar\Bank



<=====>

CUSTOMER MENU

<=====>

1. Deposit
2. Withdraw
3. Check Balance
4. Transfer
5. Change PIN
6. Mini Statement
7. Logout

<----->

Enter choice: 5

Enter current PIN: ****

Enter new PIN: ****

Please Re enter new PIN again to confirm : ****

PIN changed successfully

Press any key to continue



D:\All things\Abu Bakar\Bank



<=====>

CUSTOMER MENU

<=====>

1. Deposit
2. Withdraw
3. Check Balance
4. Transfer
5. Change PIN
6. Mini Statement
7. Logout

<----->

Enter choice: 6

Enter number of last transactions to view: 5

<=====><0.0><=====>

ID	Type	Amount	Counterparty
----	------	--------	--------------

<=====><0.0><=====>

4	TransferIn	100	10003
3	TransferOut	100	100
2	Withdraw	1232	
1	Deposit	123	

Press any key to continue|



D:\All things\Abu Bakar\Bank



<=====>

BANK ATM SYSTEM

<=====>

1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: 1

Enter account number: 10002

Enter PIN: *

Incorrect PIN. Attempts left: 2

Enter PIN: *

Incorrect PIN. Attempts left: 1

Enter PIN: *

Incorrect PIN. Attempts left: 0

Account locked due to 3 failed attempts.



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<=====>

BANK ATM SYSTEM

<=====>

1. Customer Login
2. Manager Login
3. Exit

<=====>

Enter choice: 1

Enter account number: 10002

Account locked



D:\All things\Abu Bakar\Bank



<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 1

New Account Number assigned: 10006

Enter 4-digit PIN: ****

Re-enter PIN: ****

Enter initial deposit : 100000

Account created successfully

New Account Number is: 10006

Press any key to continue|

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 6

<=====>

Display All Accounts

<=====>

1. Ascending
2. Descending

Enter choice: 1

AccNo	Balance	Status	Locked
10003	100	A	1
10005	1000	A	0
10002	2000	A	1
10001	3791	A	1
10004	15000	D	0
10006	100000	A	0

Press any key to continue



D:\All things\Abu Bakar\Bank



<=====>

MANAGER MENU

<=====>

1. Open New Account
2. Close Account
3. Deactivate/Activate
4. Unlock Account
5. Search Account
6. Display All
7. Summary Report
8. Approve Accounts
9. Back

<=====>

Enter choice: 8

<=====>

Accounts Pending Approval

<=====>

Account: 10006

Balance: 100000

Enter account number to approve : 10006

Account 10006 approved successfully

Press any key to continue

```
Microsoft Visual Studio Debug Console
=====
Thank you for using BANK ATM SYSTEM
Your money is safe
  /\_/\
 (  o.o  )
  > ^ <

Have a secure and pleasant day
=====

D:\All things\Abu Bakar\Bank ATM Manager PF Project 01\x64\Debug\Bank ATM Manager PF Project 01.
exe (process 16192) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Autom
atically close the console when debugging stops.
Press any key to close this window . . .
```